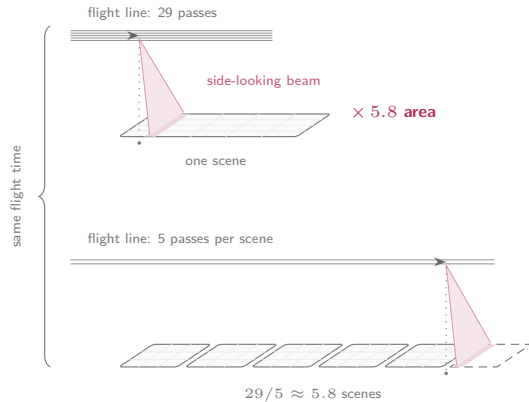
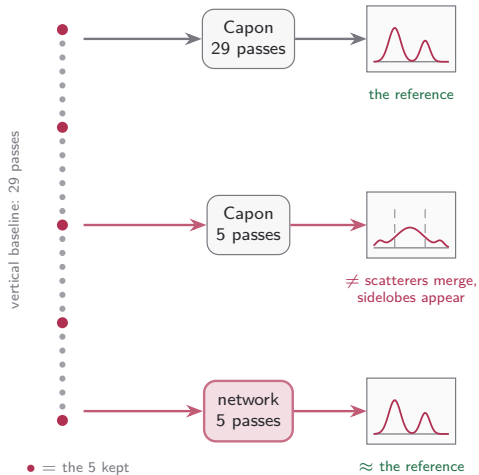


DLR-TomoSAR — Problem & amortisation

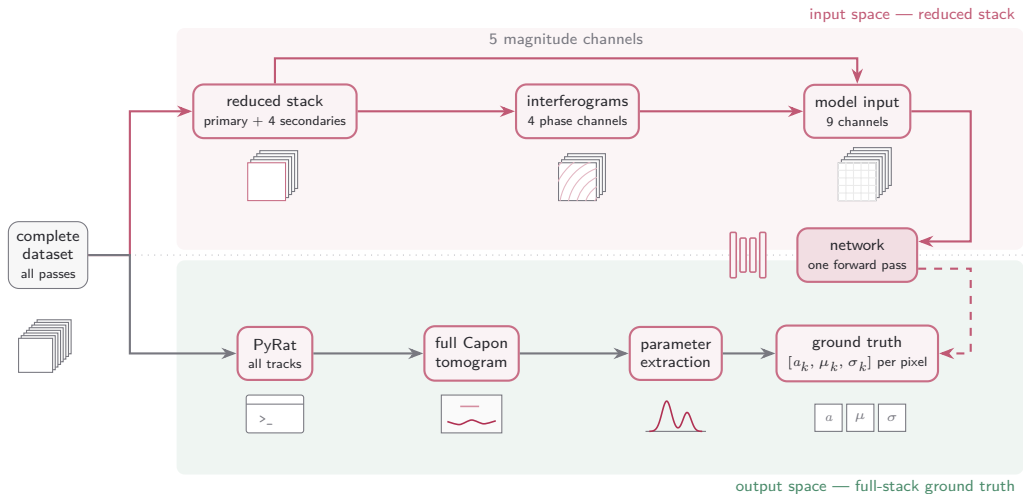
what the project does · signal model · the classical cost · amortisation

July 2026

Motivation: more area from the same flight time



What this project does



A network learns to reconstruct, from the **reduced** stack alone, the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

The tomographic signal model

Each track's baseline b_i sets its vertical wavenumber:

$$k_z^{(i)} = \frac{4\pi b_i}{\lambda r_0 \sin \theta}$$

Each pixel is a Fourier sample of the reflectivity $u(z)$:

$$g_i = \int u(z) e^{j k_z^{(i)} z} dz$$

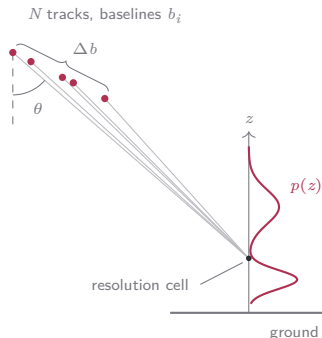
Coherences sample the spectrum of $p(z) = \mathbb{E} |u(z)|^2$:

$$\gamma_{ij} = \frac{\int p(z) e^{j (k_z^{(i)} - k_z^{(j)}) z} dz}{\int p(z) dz}$$

Classical inversion scans $\mathbf{a}(z)$ over the sample covariance $\hat{\mathbf{R}}$:

$$\mathbf{a}(z) = [e^{j k_z^{(1)} z}, \dots, e^{j k_z^{(N)} z}]^T$$

$$\hat{p}_{\text{BF}}(z) = \frac{1}{N^2} \mathbf{a}^H(z) \hat{\mathbf{R}} \mathbf{a}(z) \quad \hat{p}_{\text{Capon}}(z) = [\mathbf{a}^H(z) \hat{\mathbf{R}}^{-1} \mathbf{a}(z)]^{-1}$$



The aperture Δb sets the Rayleigh resolution

$$\delta_z = \lambda r_0 \sin \theta / (2 \Delta b).$$



Finite, irregular sampling of the Fourier axis —
tomography inverts it back to $p(z)$.

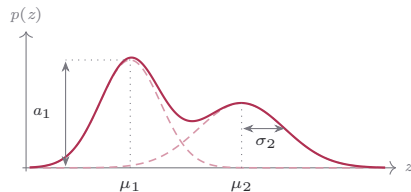
Parameterising the spectrum — and its classical cost

The elevation power spectrum at each pixel is modelled as a K -Gaussian mixture:

$$p(z) = \sum_{k=1}^K a_k \exp\left(-\frac{(z-\mu_k)^2}{2\sigma_k^2}\right)$$

a_k backscattered power of scatterer k
 μ_k its elevation in the cell
 σ_k its vertical spread

Working configuration: $K=2$ — covers 93.9% of active pixels; the $K=5$ extension for the multi-scatterer tail is in development.



Mixture $p(z)$ and its components (dashed) — three numbers per scatterer describe the whole profile.

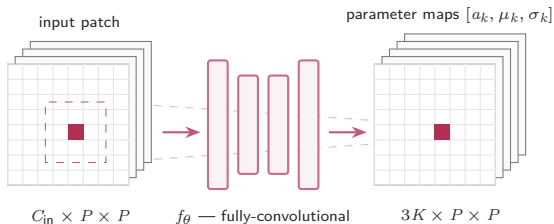


The classical route restarts an iterative fit at every pixel — the cost the amortised network removes (next).

The amortisation idea

Train a fully-convolutional network to emit all $3K$ parameters directly, for every pixel of a patch at once:

$$f_{\theta} : \mathbb{R}^{C_{\text{in}} \times P \times P} \longrightarrow \mathbb{R}^{3K \times P \times P}$$



One pixel's parameters draw on its **spatial context window** (dashed); the whole patch is decoded in a single pass.

- One forward pass replaces per-pixel iterative optimisation.
- The network sees a **spatial context window**, not an isolated pixel — whether that context matters is a question the theory does not answer (Act IV measures it).
- Supervision: parameter maps produced by the classical fit (next act) — with everything that implies about label quality (Act VI).